ASSIGNMENT 2
DIGITAL FUNDAMENTALS (BE03000071)
DEPARTMENT OF COMPUTER ENGINEERING
DIVISIONS E, F

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1) Explain the operation of an S-R flip-flop. Write the excitation table for the same.
2) Convert an S-R Flip Flop to a J-K flip-flop.
3) Give the comparison between synchronous counters and asynchronous counters.
4) Design the four-bit Johnson counter and explain the operation.
5) Design a BCD (mod 10) ripple counter using JK flip-flops.
6) Write a short note on FPGA.
7) Explain the basic structure of a CCD (Charge Coupled Device).

General Instructions:
1) The assignment should be submitted on or before 03.11.26.
2) It is to be submitted in hard copy.